

# ARCADENEXUS

Product Requirement Document (PRD)  
& brand asset specification

---

|                       |                                    |                          |                        |
|-----------------------|------------------------------------|--------------------------|------------------------|
| <b>Project Title:</b> | ArcadeNexus Online Gaming Platform | <b>Document Version:</b> | v1.0.0                 |
| <b>Author:</b>        | Ojas Sharma                        | <b>Date:</b>             | June 14, 2026          |
| <b>Category:</b>      | Web Development                    | <b>Status:</b>           | Approved / Deliverable |

# 1. PRODUCT SPECIFICATIONS & UI LAYOUT

---

ArcadeNexus is designed to serve as an interactive user-acquisition micro-portal. The platform layout is highly responsive and operates on a single-page structure consisting of five primary layout sections:

## 1.1 Navigation Bar (Header)

---

A persistent layout header containing:

- **Brand Logo:** Placed on the far left, linking to the top of the landing page.
- **Navigation Links:** Anchor references scrolling to the Game Catalog and Dashboard sections.
- **Global Volume Toggle:** An interactive sound settings control to enable or mute all audio elements.

## 1.2 Hero Section

---

The first landing viewport designed to capture user interest. It comprises:

- **Gaming Slogan:** High-impact text set in Orbitron typeface with subtle shadow glows.
- **Call-To-Action (CTA):** A prominent button triggering smooth scroll down to the Game Catalog.
- **Active User Welcome Banner:** Dynamic message greeting return players based on statistics cookies.

## 1.3 Interactive Game Catalog

---

A responsive grid container displaying game cards. Each card represents a playable game and must feature:

- Glassmorphic design overlays using transparent surface backgrounds.
- Custom thumbnail images (TTT, Snake, and Memory card references).
- Hover scaling animations with outline glows in Neon Teal.
- "Play Now" trigger button that opens the specific gameplay modal.

## 1.4 User Statistics Dashboard

---

A metrics visual section showcasing user engagement indicators read from local state storage. Includes tabular views for high scores, win rates, and total played counters, alongside a button to wipe cached stats.

## 1.5 Platform Footer

---

Standardized footer containing copyright info, developer credits, and references to asset licensing agreements.

## 2. TECHNICAL SPECIFICATIONS & GAMEPLAY ENGINES

---

The platform must be developed using a pure frontend architecture (HTML5, Vanilla CSS3, and ES6+ Javascript) without compilation steps or third-party framework runtime imports.

### 2.1 Games Engineering Logic

---

#### 1. Tic-Tac-Toe:

- Fires inside a 3x3 clickable grid.
- Player VS local AI agent. The AI must check board positions to perform basic blocks and attacks.
- Checks grid states after every turn to declare victory lines or draw status.

#### 2. Snake:

- Rendered using an HTML5 Canvas drawing loop.
- Control interface binds keyboard arrows and WASD buttons.
- Increases move speed by 5% increments every time the score grows by 5 food items.
- Self-collision or boundary collision terminates the loop and fires game-over updates.

#### 3. Memory Match:

- Loads a shuffled 4x4 layout containing 8 card pairs.
- CSS transitions trigger realistic card flip motions.
- Mismatched cards flip back face down after a 1.0 second delay timeout.
- Keeps count of total flip attempts to record moves metrics.

### 2.2 Local State Persistence Schema

---

All metrics must be loaded and saved via the browser's native `localStorage` utilizing a single structured key named `arcadenexus_stats` using this schema:

```
{
  "total_games_played": 15,
  "games": {
    "tictactoe": { "played": 5, "wins": 3, "losses": 1, "ties": 1, "high_score": 3 },
    "snake": { "played": 6, "wins": 0, "losses": 6, "ties": 0, "high_score": 142 },
    "memory": { "played": 4, "wins": 2, "losses": 2, "ties": 0, "high_score": 12 }
  }
}
```

### 3. VISUAL & AUDITORY BRAND ASSETS SPECIFICATION

Developers must implement and reference only the assets listed below. Custom color injection or unlisted asset imports will breach acceptance criteria.

#### 3.1 Brand Color Guidelines

|                                                                                   |                                                                                   |                                                                                    |                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |
| Deep Void<br>#0B0C10                                                              | Cyber Gray<br>#1F2833                                                             | Neon Teal<br>#66FCF1                                                               | Laser Cyan<br>#45A29E                                                               |

#### 3.2 Graphic Assets Inventory

| Asset ID    | Relative Location             | File Type  | Description                                  |
|-------------|-------------------------------|------------|----------------------------------------------|
| logo_full   | assets/images/logo.svg        | SVG Vector | Primary vector brand identity emblem.        |
| card_ttt    | assets/images/ttt_card.png    | PNG Image  | Cover image for Tic-Tac-Toe selection card.  |
| card_snake  | assets/images/snake_card.png  | PNG Image  | Cover image for Snake selection card.        |
| card_memory | assets/images/memory_card.png | PNG Image  | Cover image for Memory match selection card. |

#### 3.3 Auditory Assets Inventory

| Asset ID       | Relative Location            | Format            | Playback Context                         |
|----------------|------------------------------|-------------------|------------------------------------------|
| sound_click    | assets/audio/click.mp3       | MPEG Audio (.mp3) | Button hovers and selection clicks.      |
| sound_score    | assets/audio/score.mp3       | MPEG Audio (.mp3) | Point points or memory card matches.     |
| sound_gameover | assets/audio/gameover.mp3    | MPEG Audio (.mp3) | Match completions and game-over updates. |
| music_ambient  | assets/audio/cyber_beats.mp3 | MPEG Audio (.mp3) | Looped background soundtrack.            |